APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular & Supplementary Exam December 2020

Course Code: EE367

Course Name: NEW AND RENEWABLE ENERGY SYSTEMS

PART A

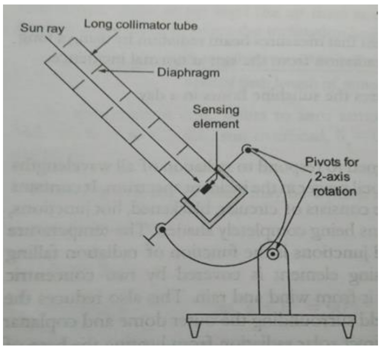
1) Explain the importance of non-conventional energy resources. Mention their advantages.

Non-conventional resources are those energy resources which can be used again and again. It means renewable resources are those which can be reused without much efforts and energy can be received at much lower cost. Such resources never ended. Hydro Energy, Solar Energy, Geo-thermal Energy, Tidal Energy, Bio-Energy are a few of renewable resources.

- There are a lot of advantages of non-conventional energy resources. A few of them are:
- No Fuel Cost: Renewable energy resources provide energy continuously at negligible fuel cost and hence these are cheap source of energy which makes them favourable over other sources of energy.
- Pollution Free and Eco-Friendly: These Non-Conventional Energy Resources are pollution free and hence are Eco-friendly. This is one of the best things to be considered in this age of global warming.
- Simple Design: Renewable energy sources have simple plant design and hence easy to operate and no specialized workforce is requires in these plants.
- Lower maintenance cost: Being simple in design the maintenance cost of these plants is very less and these plants can thus produce electricity at much lower cost than other plants.
- Available at load center: These plants can be installed remotely and at locations completely off grid. Hence better choice for single plant use.

2) Draw the schematic diagram of Pyrheliometer and explain its working.

- A Pyrheliometer is an instrument for measurement of direct beam solar irradiance. Sunlight enters the instrument through a window and is directed on a thermopile which converts heat to an electrical signal that can be recorded. The signal voltage is converted via a formula to measure watts per square meter. It is used with a solar tracking system to keep the instrument aimed at the sun. Sunlight enters the instrument through collimator tube and is directed onto a thermopile (sensing element) which converts heat to an electrical signal that can be recorded. Uses a long collimator tube to collect beam radiation whose field of view is limited to a solid angle of 5.5° by appropriate diaphragms inside the tube. Inner surface is blackened to absorb any radiation. At the base of tube a wire wound thermopile having a sensitivity of approximately 8 micro volt/W/M^2. Tube sealed with dry air to eliminate absorption of beam of radiation within the tube by water vapour



3) What are the factors that affect the efficiency of solar cell? Explain.

   1. SHADE:

   As we know that electrical power is produced by sunlight by converting it. If something comes between the plates and sunlight then it is obvious that it is going to affect the production. Even small shades can reduce a good amount of production.

   2. STATE OF CABLE & WIRING SYSTEM

   Common voltages for Solar PV (DC) systems are around 12V, 24V to 48V. Solar panels connected in series provide more voltage. While solar panels connected in parallel yield more current. Electrical appliances usually require 220 volts of voltage.

   3. ORIENTATION:

   For better efficiency, your solar cells should be positioned in the direction where they can absorb radiation and rays better. If your cells are facing west or east, they would produce 20% lesser energy if they were facing south.

   4. TEMPERATURE:

   Temperature is one of the biggest factors that can increase and decrease solar cell efficiency. Increasing in temperature more than the set limit can affect in negative. If the temperature of solar cells rises above 25 degrees celsius, the net output decreases with the rise of temperature.

   5. MAINTENENCE:

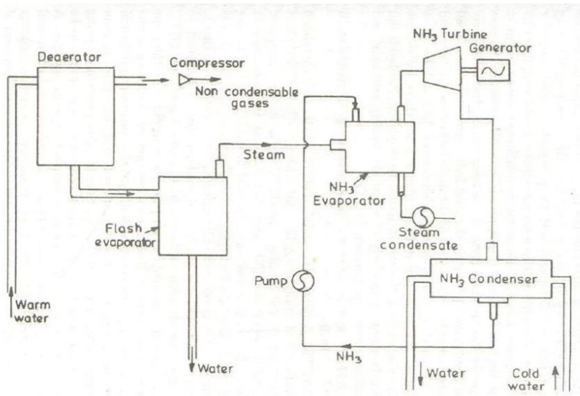
   Solar cells need little maintenance per year especially if the system is grid-tied however it is recommended to wash cells routinely as dirt & dust affect the efficiency of solar panels.

6. THE EFFICIENCY OF THE BATTERY

Batteries play an important role for backup in a Solar PV system. The efficiency of an entire system depends on all of the components.

4) Draw the block diagram of a hybrid cycle OTEC system and mention its advantages.

The last cycle proposes to combine both open and closed cycles. The warm water is vaporized in a vacuum chamber. The steam transforms working liquid into vapour (heat exchanger) which is directed in a turbine. As in the open-cycle the condensed water provides fresh water. There is an attempt to combine the best futures of Open & Closed Cycle OTEC. Sea Water is Flash evaporated to steam. Heat resulting steam is then transferred to ammonia.



Advantages

- Reduction in the dependence on fossil fuels.
- Source of energy is free, renewable and clean.
- Clean electricity is produced with no production of greenhouse gas or pollution (Liquid or solid).
- Energy produced is free once the initial costs are recovered.
- These technologies are renewable sources of energy. OTEC systems can produce fresh water as well as electricity. This is a significant advantage for an island, such as the Virgin Islands for example, where fresh water is limited.
- There is enough solar energy received and stored in the warm tropical ocean surface layer to provide most, if not all, of present human energy needs.

5) Discuss the advantages and disadvantages of wind energy conversion systems.

Advantages

- Air as a fuel is free and inexhaustible.
- It is clean source of energy and does note pollute the environment.
- The cost of electricity is too low and wind turbine could be used over more than 20 years.

- It's cheap as only the installation and maintenance cost is required.
- Wind energy is one of the fastest growing sectors all around the globe so it is generating a lot of employment in manufacturing, installation and maintenance.

Disadvantages

- It takes a lot of research and effort to decide the location where wind power plant has to be installed, due to fluctuating pattern of wind.
- Its initial setup cost is too high as to setup a turbine you have to go through a survey to determine the wind speed of the location. It all adds up to the cost.
- They are the greatest disadvantage to local bird population as they die due to collision with blades.
- Noise pollution is the one of the major disadvantage.
- Wind power plant is only useful to the countries with coastal or hilly areas.

6) Write a short note on wind and its properties.

The wind is used to generate mechanical power. Wind turbines convert the kinetic energy in the wind into mechanical power. This mechanical power can be used for specific tasks (such as grinding grain or pumping water) or a generator can convert this mechanical power into electricity. Wind is a form of solar energy. Winds are caused by the uneven heating of the atmosphere by the sun, the irregularities of the earth's surface, and rotation of the earth. Wind flow patterns are modified by the earth's landscape, bodies of water, and vegetation. Humans use this wind flow, or motion energy, for many purposes: sailing, flying a kite, and even generating electricity.

Wind: K.E associated with movement of large masses of air

- Features: clean, cheap, eco-friendly renewable
- Wind energy Mechanical Electrical
- Moderate wind of 5m/s-25m/s favourable
- Energy generation cost Rs.2.5/KWh
- Installation cost –Rs.4 Cr/MW-comparable with thermal power plant.
- Energy payback period: 1yr
- Expected lifetime of wind turbine: 20 Yr

7) Describe the process of biomass to ethanol conversion.

- Ethanol is manufactured by the action of microorganisms on carbohydrates. The process is known as *alcoholic fermentation.*
- *Carbohydrates* (also known as saccharids) can be divided into three major classes in order of increasing complexity.

(i) Monosaccharides : They are simple hydrocarbons which cannot be hydrolyzed into simpler compounds. Eg: glucose (C6 H12 O6).

(ii) Oligosaccharides : These yield around 2-10 of monosaccharide molecules on hydrolysis.

(iii) Polysaccharides : These are high molecular mass carbohydrates, which yield a large of monosaccharides molecules on hydrolysis.

- The hexose (glucose and fructose) required for ethanol fermentation is derived from :

1. **From Sucrose** : most common disaccharide and is manufactured from sugar cane or beetroot.

- Usually , commercial sucrose is removed from cane sugar and the remaining molasses is used for ethanol production.

$$C_{12}H_{22}O_{11}\text{ (Sucrose)} \xrightarrow{\text{Hydrolysis}} C_6H_{12}O_6\text{ (Glucose)} + C_6H_{12}O_6\text{ (Fructose)}$$

2. **From Starch:** On hydrolysis with dilute H2SO4 or enzymes, starch breaks down to maltose and finally to glucose.

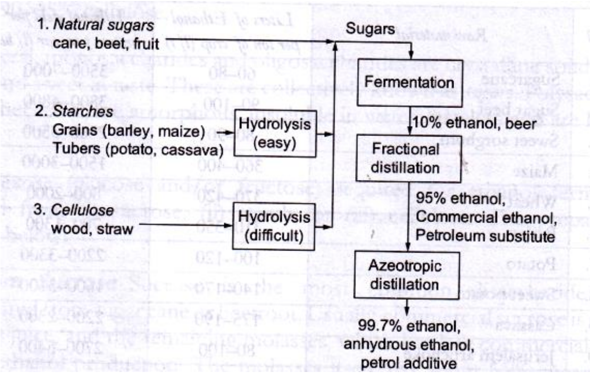
$$2(C_6H_{10}O_5)_n\text{ (Starch)} + nH_2O \longrightarrow nC_{12}H_{22}O_{11}\text{ (Maltose)} + nH_2O \longrightarrow 2nC_6H_{12}O_6\text{ (Glucose)}$$

3. **From Cellulose :** It is not hydrolyzed easily, but on heating with H2SO4 under pressure yields glucose

$$(C_6H_{10}O_5)_n\text{ (Cellulose)} + nH_2O \longrightarrow nC_6H_{12}O_6\text{ (Glucose)}$$

- Finally, ethanol is obtained from fermentation of hexose sugars.

$$C_6H_{12}O_6\text{ (hexose)} \xrightarrow[\text{Fermentation}]{32^\circ C} 2C_2H_5OH\text{ (ethanol)} + 2CO_2$$

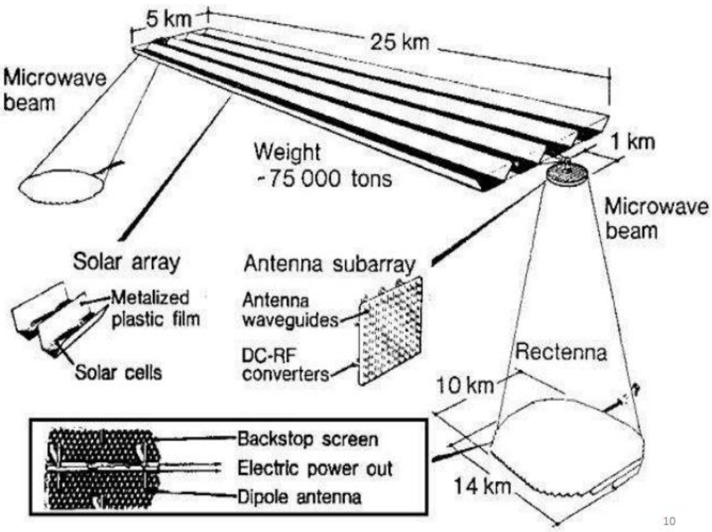


8) "Satellites can be used for energy harvesting". How?

Satellites harness power from solar panels to power the spacecraft. This high voltage supply needs to be stored and converted for all the on board electronics. Our large portfolio of space qualified devices and reference designs help you maximize your system performance.

Modern Satellite Power Distribution systems require:

- Multiple low noise and efficient point of load supplies.
- Efficient Isolated power supplies to convert the main bus to point of load supplies.
- A feedback method to monitor the secondary supply rail.



SATELLITE POWER SYSTEM

ADVANTAGE:

- Unlimited energy source
- Unlike coal and nuclear fuels SPS does not require mining operation
- Energy delivery anywhere in the world
- Zero fuel cost
- Zero emission of CO2 and other hazardous waste
- Efficient collection of solar radiation
- Will not prone to the threat of terrorist like nuclear power plant

DISADVANTAGE:

- Storage of electricity during off peak demand hour
- Frequency of beamed radiation is planned to be at 2.45GHz and this frequency is used by communication satellite also
- Massive structure
- High initial cost and much time for construction
- Requires a network of hundred of satellite
- Size of spacetenna and rectenna

13

PART B

9 a) Explain different methods used for energy storage.

Basically Classified as:

- Battery Storage Systems – Lithium-ion & Redox flow batteries
- Thermal Systems – Compressed air storage, liquid air storage, pumped heat electrical storage & methane
- Mechanical Systems – Pumped storage hydroelectricity, Advanced rail storage, Flywheel energy storage
- Super conducting magnetic storage

Thermal Storage Systems

- Thermal systems use heating and cooling methods to store and release energy.
- For example, molten salt stores solar-generated heat for use when there is no sunlight.

- Ice storage in buildings reduces the need to run compressors while still providing air conditioning over a period of several hours.
- Other systems use chilled water and dispatchable hot water heaters.
- In all cases, excess energy charges the storage system (heat the molten salts, freeze the water, etc.) and is later released as needed.

Liquid Air Storage:

- Liquid Air Energy Storage, or LAES, uses excess grid electricity to cool ambient air to the point it becomes a liquid.
- To extract electricity from the system, the liquid air is converted back to gas by exposure to ambient air or with waste heat.
- This expanding gas is then used to power turbines.

Batteries

- There are various forms of batteries, including: lithium-ion, flow, lead acid, sodium, and others designed to meet specific power and duration requirements.
- Lithium-ion batteries now have a range of applications including smaller residential systems and larger systems that can store multiple megawatt hours (MWh) and can support the entire electric grid.
- These systems typically house a large number of batteries together on a rack, combined with monitoring and management units.
- These systems have a small footprint for the amount of energy they store.
- For example, a system the size of a small refrigerator could power an average home for several days.
- A utility-scale system of 100 MWh could fit on less than 0.5 acres.
- Lithium-ion batteries have received a lot of press for their rapidly declining costs, due to the growing popularity of electric vehicles.
- A different type of battery is a flow battery in which energy is stored and provided by two chemicals that are dissolved in liquids and stored in tanks. These are well suited for longer duration storage.

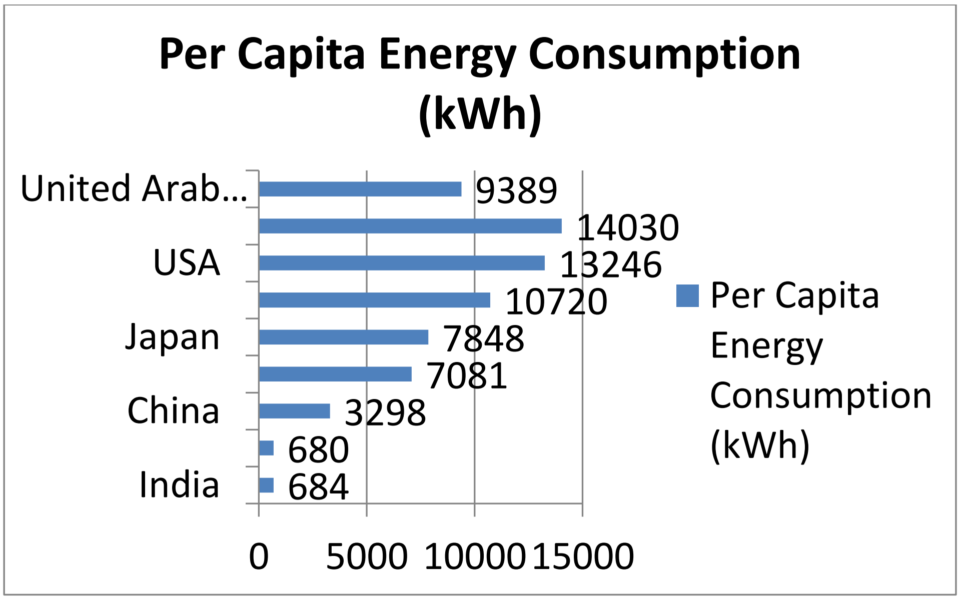
9 b) Compare non-concentrating and concentrating solar collectors.

TABLE 3.1 Differences between flat and focussing collectors

Flat collectors	*Focussing collectors*
The absorber area is large	Absorber area is small
Concentration ratio is 1	Concentration ratio is high varying from 4 to 3000
Temperature range is low, generally not more than 70°C	Tmperature range is high, which is up to 3000°C
It uses both beam and diffuse radiation	It uses mainly beam radiation
Simple in construction and maintenance, no tracking system is required	More complicated design and difficult maintenance, tracking system is required
Less costly	More costly
Application limited to low temperature uses	High temperature applications such as power generation
Suitable for all places as it can work in clear and cloudy days	Suitable where there are more clear days in a year

10 a) Write short notes on global energy scenario.

- ☐ There is uneven pattern of energy consumption in different country
- ☐ Developed country have a population of 10%, use approx. 90% of the available resources
- ☐ On the other side, there is country where people depend on wood as a form of energy
- ☐ From last 20 year developing country focus on energy sector
- ☐ Rate of growth in energy sector in developed country is 1% to 2%, but in developing country growth rate is 7 to 8%.

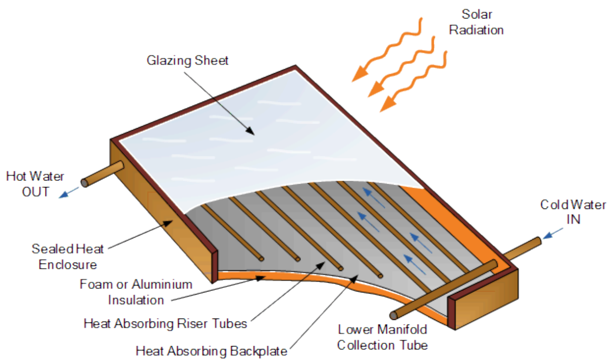


- ☐ Global Energy requirement increases continuously
- ☐ Fuel resources are depleted
- ☐ Oil price increase continuously
- ☐ It put pressure on country economic
- ☐ Oil war start (developed country pressurized oil rich country)
- ☐ It is estimated that in next 50-60 year oil & gas resources are consumed

10 b) Explain in detail the stages of heat transfer process from solar radiation to the heat absorbing liquid medium in a flat plate collector.

- Heat conversion by green house effect.
- Energy from the sun comes in the form of light, a shortwave radiation. When this radiation strikes a solid or liquid, it is absorbed and transformed into heat energy, the material become warm and stores the heat, conducts it to surrounding materials (water, air, other solids or liquids) or reradiates into other materials of lower temperature. This radiation is a long wave radiation.

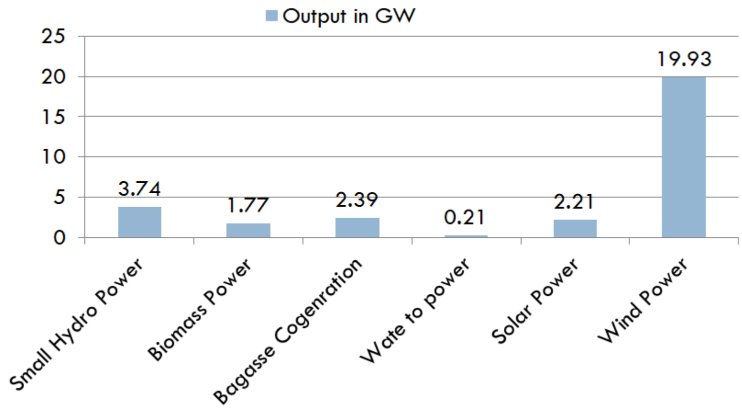
- Glass easily transmits short wave radiation, which means that it poses little interference to incoming solar energy, but it is a very poor transmitter of long wave radiation.
- Once the sun's energy has passed through the glass windows and has been absorbed by some material inside, the heat will not be reradiated back outside.
- Glass act as heat trap.
- A black painted plate absorbs the incoming sunlight. About it, is fixed a plate of ordinary window glass. When the temperature of the black plate increases, it emits an increment of thermal heat in the form of infrared light.
- Flat-Plate collectors are the most common solar collectors for use on solar water- heating systems in homes and in solar space heating.
- A Flat-plate collector consists basically of an insulated metal box with a glass or plastic cover (the glazing) and a dark-coloured absorber plate.
- Solar radiation is absorbed by the absorber plate and transferred to a fluid called Glycol, that circulates through the collector in tubes.
- Flat-Plate collectors heat the circulating fluid to a temperature considerably less than that of the boiling point of water and are best suited to applications where the demand temperature is 30-70 degrees
- Flat-plate solar collectors may be divided into two main classifications based on the type of heat transfer fluid used.
- <u>Liquid heating collectors</u> are used for heating water and non-freezing aqueous solution and occasionally for non-aqueous heat transfer fluids.
- <u>Air or gas heating collectors</u> are employed as solar air heaters.



11 a) Give an overview of non-conventional energy potential and utilization of different states of India

- Targeted renewable power generation by 2022: 175 GW
- Targeted non-fossil fuel based installed electric capacity by 2030: 40%
- Solar power capacity increased by more 14 times in the last 5 years
- India is a seventh largest country and have a population of 1.2 billion people
- To maintain growth rate, need rapid growth in energy sector
- 41% of electricity generation from thermal power plant

- By 2016-2017, total domestic energy production of 670 million tons of oil equivalent (MTOE). This meet only 71% of the expected demand.
- As per the 2011 Census, 55.3% rural households had access to electricity
- Still most of the rural area have limited supply hours of electricity
- India ready to exploit renewable energy resources
- Ministry of New and Renewable Energy (MNRE) come in picture in 2006
- It work to increase the share of renewable energy



Region	Power (MW)			
	Peak Demand		Peak Met	
	Jan 2018	Jan 2019	Jan 2018	Jan 2019
Northern	47214	47925	46252	47356
Western	50477	53698	50085	53505
Southern	43306	44560	43115	44560
Eastern	18526	19641	18256	19523
North Eastern	2339	2575	2317	2552
All India	158640	163224	156720	162349

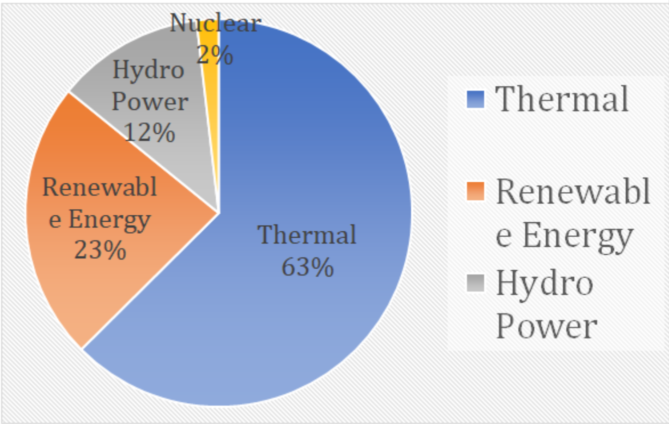
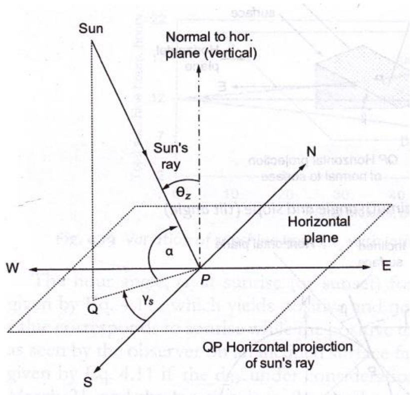


Fig.1: Installed Power Generation Capacity (368 GW)

11 b) Explain the following i) inclination angle ii) zenith angle iii) solar azimuth angle iv) tilt angle iii) angle of incidence.



<u>Inclination or altitude angle(α)</u>

- It is the angle between sun's ray and its projection on horizontal surface.

<u>Zenith angle(θz)</u>

- It is the angle between sun's ray and perpendicular (normal) to the horizontal plane.

<u>Solar azimuth angle(γs)</u>

- It is the angle on a horizontal plane, between the line due south and the projection of sun's ray to the point on the horizontal plane. It is taken as positive when measured from south towards west.

<u>Tilt or slope angle(β)</u>

- It is the angle between the inclined plane surface, under consideration and the horizontal. It is taken to be positive for the surface sloping towards south.

<u>Surface Azimuth angle(γ)</u>

- It is the angle in horizontal plane, between the line due south and the horizontal projection of normal to the inclined plane surface. It is taken as positive when measured from south towards west.

<u>Angle of incidence(θ)</u>

- It is the angle between sun's ray incident on the plane surface (collector) & normal to that surface.

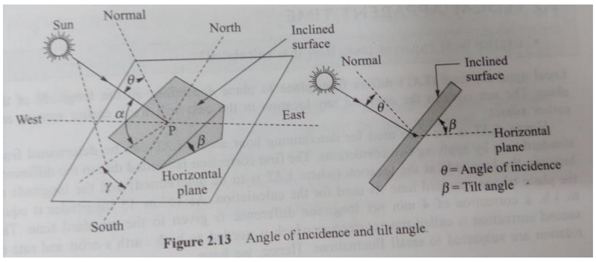
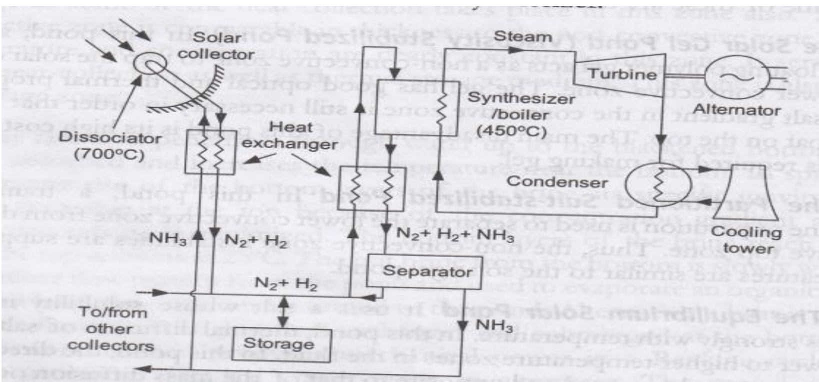


Figure 2.13 Angle of incidence and tilt angle

PART C

12 a) Explain the working of a solar thermal electric power plant with neat sketches.



- **Solar thermal energy** (**STE**) is a technology for harnessing solar energy for thermal energy (heat) requirement in industries, residential sector and commercial setup.
- S**olar Hot Water Systems**, also known as **Solar Thermal Systems (STS)** operate differently from photovoltaic (PV) systems.
- Solar thermal systems run water or fluid through their system. The solar panels are heated and the water running through them gets heated. The heated water is ready for use.
- Solar thermal technologies capture the heat energy from the sun and use it for heating and/or the production of electricity.
- This is different from photovoltaic solar panels, which directly convert the sun's radiation to electricity.
- There are two main types of solar thermal systems for energy production – active and passive.

- Active systems require moving parts like fans or pumps to circulate heat-carrying fluids. Passive systems have no mechanical components and rely on design features only to capture heat. (e.g. greenhouses)

Solar thermal technologies are designed to convert the incident solar radiation into usable heat. The process of solar heat conversion implies using energy collectors - the specially designed mirrors, lenses, heat exchangers, which would concentrate the radiant energy from the sun and transfer it to a carrier fluid. The fluid passes through the sunlight collector and becomes very hot. Typical heat carrier fluids are water/steam, oil, or molten salt. Then the fluid is transferred to the heat engine, which converts the heat to electricity.

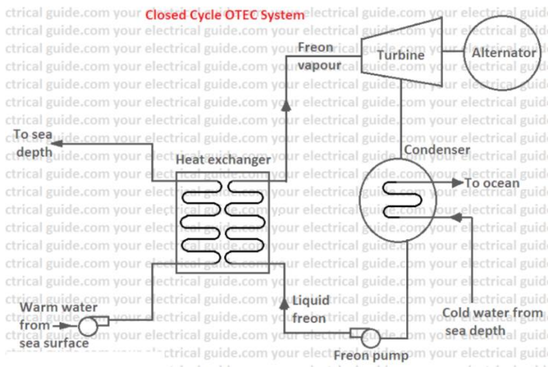
The schematic diagram of a typical distributed collector power generation plant is shown in Fig. 5.39. The heat collected in collectors is used to dissociate ammonia into nitrogen and hydrogen at high pressure (approx. 300 atm). The heat of reaction, 46 kJ/mole (of NH_3) is provided by the solar energy. This nitrogen–hydrogen mixture is transported to the central plant where N_2 and H_2 are recombined in a synthesizer, using a catalyst. The heat released during the reaction is utilized in a heat engine to generate electric power through an alternator. In the synthesizer only a part of nitrogen and hydrogen recombine to produce ammonia. The products of the synthesizer are cooled to liquefy ammonia, separate it out and send it to the collector system. The mixture of nitrogen and hydrogen that remains in gaseous state is fed back to the synthesizer.

12 b) Describe the Principle of operation of tidal energy conversion process.

- **Tidal Energy** or **Tidal Power** as it is also called is another form of hydro power that utilizes large amounts of energy within the oceans tides to generate electricity.
- Earth uses the gravitational forces of both the moon and the sun every day to move vast quantities of water around the oceans and seas producing tides.
- As the Earth, its Moon and the Sun rotate around each other in space, the gravitational movement of the moon and the sun with respect to the earth, causes millions of gallons of water to flow around the Earth's oceans creating periodic shifts in these moving bodies of water.
- These vertical shifts of water are called "tides".
- When the earth and the moons gravity lines up with each other, the influences of these two gravitational forces becomes very strong and causes millions of gallons of water to move or flow towards the shore creating a "high tide" condition.
- The potential energy of the water contained in the daily movement of the rising and falling sea levels to generate electricity.
- The water flows in and out of the turbines in both directions instead of in just one forward direction.

- Tidal energy, just like hydro energy transforms water in motion into a clean energy.
- The motion of the tidal water, driven by the pull of gravity, contains large amounts of kinetic energy in the form of strong tidal currents called tidal streams.
- The daily ebbing and flowing, back and forth of the oceans tides along a coastline and into and out of small inlets, bays or coastal basins, is little different to the water flowing down a river or stream.

13 a) Draw the schematic and explain the working of a closed cycle OTEC plant.



- This cycle uses a working fluid (with a low boiling point) which is cooled down and heated up in a full cycle.
- The warm water located at the surface is injected into a heat exchanger; the working fluid gets the heat to change its state, and is vaporized.
- The vapour is directed through the turbine and generator where the electricity is produced.
- Then the vapour is condensed by the deep cold water.
- The liquid goes back to the heat exchanger and so on.
- India has built a closed-cycle, floating plant of 1 MW rated power.
- It requires a special working fluid that receives & rejects heat to the source to sink.
- Working fluid may be ammonia, propane or Freon. Operating pressure of this fluid is 10Kg/Cm2 (10Bar).
- Specific volume is much lower to steam in conventional power plant.
- This leads to smaller size turbine, when compared to Open cycle OTEC turbine.
- Closed cycle system is to transfer heat efficiently across the heat exchanger surfaces and uses very large heat exchangers having the typical efficiency of 2%.
- In this system, working fluids having a low boiling point like ammonia, Freon-12, butane gas are used for heat engine because the working temperatures of sea water are small.

- The warm water from the surface of the sea is circulated through a heat exchanger with help of a pump.
- In the heat exchanger, the heat of seawater is absorbed by Freon and Freon vapors are generated at high pressure.
- This vapor expands in the turbine to drive it and a coupled generator with it.
- The Freon vapor from the turbine condensed in a condenser with the help of cold water.
- The Freon condensate is pumped again into the heat exchanger and the complete cycle is repeated.
- The overall efficiency of such a plant is very low in the range of 2 to 3% only.

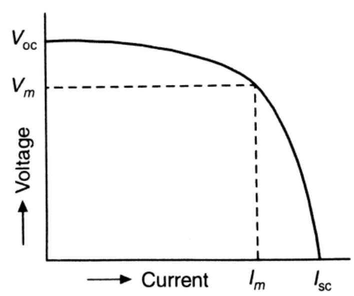
13 b) Define the following terms associated with solar PV i) Fill factor ii) Conversion efficiency iii) Maximum power point iv) Poly crystalline cell

• **FF = fill factor** – The fill factor is the relationship between the maximum power that the solar cell can actually provide under normal operating conditions and the product of the open-circuit voltage & the short-circuit current, (Voc x Isc) This fill factor value gives an idea of the quality of the array and the closer the fill factor is to 1 (unity), the more power the cell can provide. Typical values are between 0.7 and 0.8.

$$\mathbf{FF} = \frac{\mathbf{V_{mp} I_{mp}}}{\mathbf{V_{oc} I_{sc}}}$$

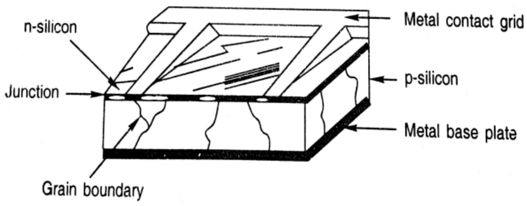
- **%eff = percent efficiency** – The efficiency of a photovoltaic array is the ratio between the maximum electrical power that the array can produce compared to the amount of solar irradiance hitting the array. The efficiency of a typical solar array is normally low at around 10-12%, depending on the type of cells (monocrystalline, polycrystalline, amorphous or thin film) being used.

• **MPP = maximum power point** – This relates to the point where the power supplied by the cell that is connected to the load is at its maximum value, where MPP = Imp x Vmp. The maximum power point of a photovoltaic is measured in Watts (W) or peak Watts (Wp).



In polycrystalline solar cell, *liquid silicon* is used as raw material and polycrystalline silicon was obtained followed by *solidification process*. The materials contain various crystalline sizes. Hence, the efficiency of this type of cell is less than Mono-crystalline cell.

- Its efficiency is 13-15 %.
- It is made up of large number of different silicon crystal.
- Compared to mono-crystalline it is less expensive.
- The poly crystalline silicon is in the form of granules and the size of crystallites manly depends on the cooling condition
- The silicon cells made from polycrystalline silicon are of low cost and low efficiency



Cross-section of polycrystalline silicon solar cell

14 a) Compare the components and working of a standalone and grid connected PV system.

Stand-alone PV system:

It is located at the load center and dedicated to meet all the electric loads of a village or a specific set of loads.

- Energy storage is generally essential.
- It is more relevant and successful in remote and rural areas having no access to grid supply.

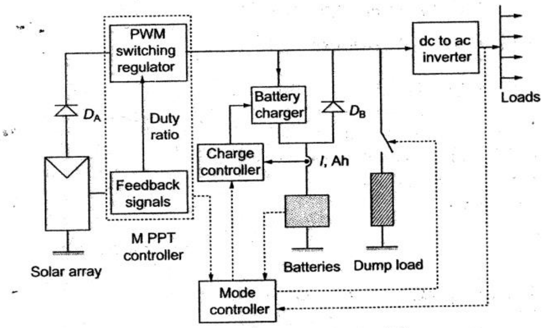


Fig. 6.26 A general stand-alone solar PV system

- MPPT senses the voltage and current outputs of the array adjust the operating points to extract maximum power under the given climatic conditions.
- Output of the array after converting to AC is fed to loads.
- The array output in excess of load requirement is used to charge the battery. If excess power is still available after fully charging the battery it may be shunted to dump heaters.

- If sun is not available battery supplies the load through an inverter .
- Battery discharge diode Db prevents the battery from being over charged after the charger is opened.
- The array diode Da is to isolate the array from the battery to prevent battery discharge through array during nights.
- A mode controller is a central controller for the entire system.
- It collect the system signal and keeps the track of charge or discharge state of battery, matches generated power and loads and commands the charger and dump heater on-off operation.

Grid Interactive system:

- This system is connected to utility grid with two way metering system.
- It may be a small roof top system owned and operated by the house owner or a relatively bigger system meant for whole village or a community.
- It meets day time requirements of the house owner without any battery backup and surplus power is fed to the grid.
- During peak hours and during night's energy shortage may be met from grid.

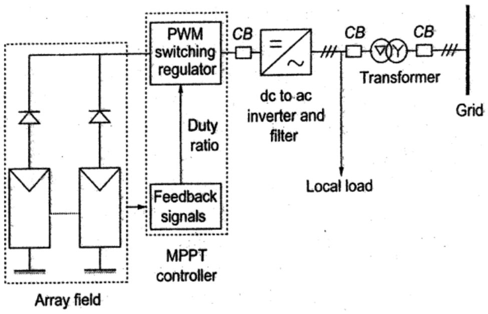


Fig. 6.27 A general grid-interactive solar PV system

- Here all excess power is fed to grid and dump heaters are not required.
- During absence of sunshine supply of power is maintained from the grid and thus battery is eliminated.
- Recently PV modules are being made with inverters as an integral component in the junction box of modules which is known as AC PV modules.

14 b) Explain the characteristics of a site suitable for the construction of a tidal power plant. List any two advantages and limitations of a tidal power plant.

1. Availability of Raw-materials
2. Nearness to the Market
3. Nearness to Sources of Operating Power
4. Labour Supplies
5. Transportation
6. Finance
7. Climate
8. Industrial Inertia

Advantages:

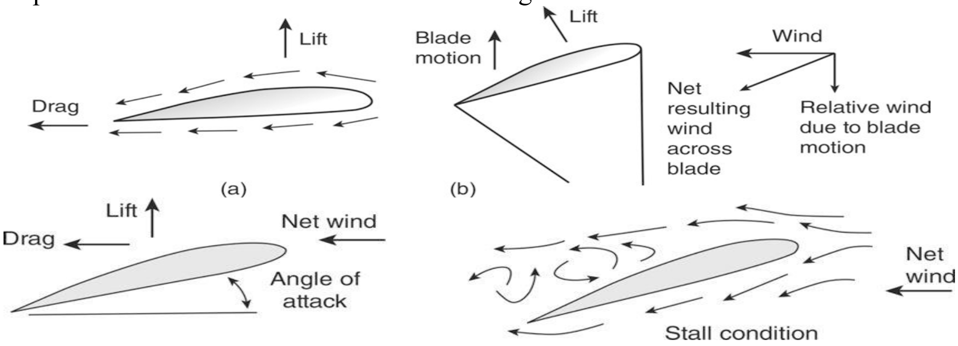
- Tidal energy is a renewable energy resource because the energy it produces is free and clean as no fuel is needed and no waste bi-products are produced.
- Tidal energy has the potential to produce a great deal of free and green energy.

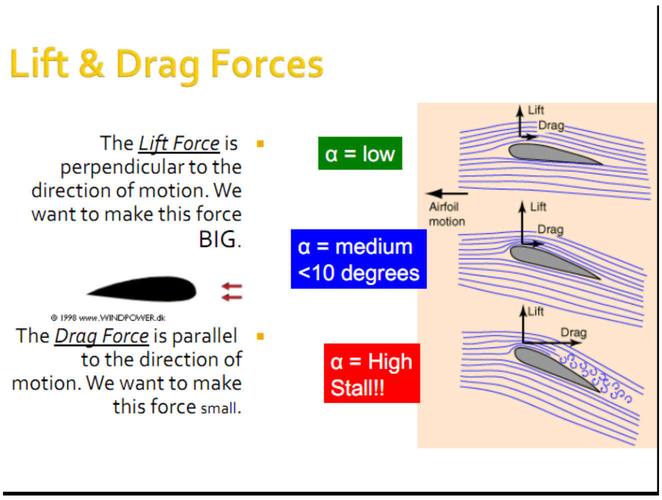
Disadvantages:

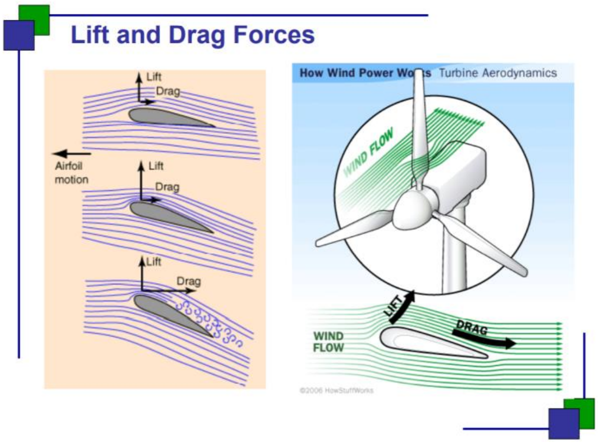
- Tidal energy is not always a constant energy source as it depends on the strength and flow of the tides which themselves are effected by the gravitational effects of the moon and the sun
- Tidal Energy requires a suitable site, where the tides and tidal streams are consistently strong

PART D

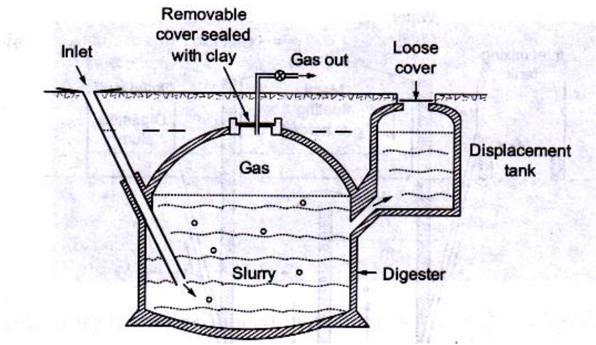
15 a) What is meant by lift and drag forces acting on wind turbine. Explain the importance of these forces in wind turbine design.







15 b) Explain the construction and working of fixed dome type biogas plant with the help of a figure.

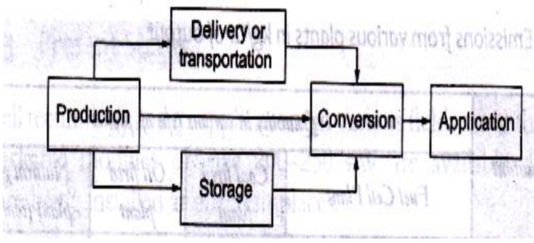


- It is built either of cement-concrete or brick-walled.
- The gas is stored in the upper dome. With increase in gas, pressure in dome rises up.
- It pushes of the digested material out of the tank.
- The costs of a fixed-dome **biogas plant** are relatively low.
- These plants are more economical as only masonry work is required.
- Gas pressure in the dome varies depending on the production/consumption rate.
- By construction, a dome structure is very strong for outside pressures but a weak one for inner pressures.
- As gas pressure is exerted from inside out, the dome structure may fail if proper care is not taken in its construction.
- Therefore skilled masonry workmanship is required for construction of dome. also., in case of any leakages/cracks, the plant may fail
- The slurry enters from an inlet & the digested slurry is collected in a displacement tank
- Stirring is required if the raw material is crop residue
- There is no bifurcation in the digester chamber & therefore the gas production is somewhat less as compared to a floating point design.
- The gas produced is stored in the dome & displaces the liquids in inlet & outlet, often leading to gas pressures as high as 100 cm of water
- The gas occupies about 10% of the volume of the digester.

- As the complete plant is constructed underground, the temperature tends to remain constant & is often considerably higher than the ambient temperature in winter.

16 a) What are different methods used for the production and storage of hydrogen.

- Hydrogen holds the potential to provide clean, reliable and affordable energy supply that can enhance economy, environment and security.
- It is non – toxic and recyclable.
- H2 can be produced by using a variety of energy sources , such as solar, nuclear and fossil fuels and can be converted to useful energy forms efficiently and without detrimental environmental effects.
- When burned as fuel or converted to electricity it joins with O2 to produce energy with water as the only emission.
- Despite all this, realization of H2 economy faces multiple challenges. Unlike natural gas, gasoline H2 has no existing, large scale supporting infrastructure.

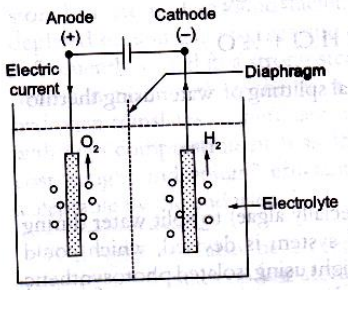


- It can be produced through two routes:
    a. Fossil fuels such as natural gas, coal, methanol, gasoline etc are decomposed by thermo-chemical methods to obtain hydrogen.
    b. It can also be produced by splitting water into hydrogen and oxygen by using energy from nuclear or renewable sources such as solar, wind, geothermal through electrical or thermal means. (ie, electrolysis and thermolysis)

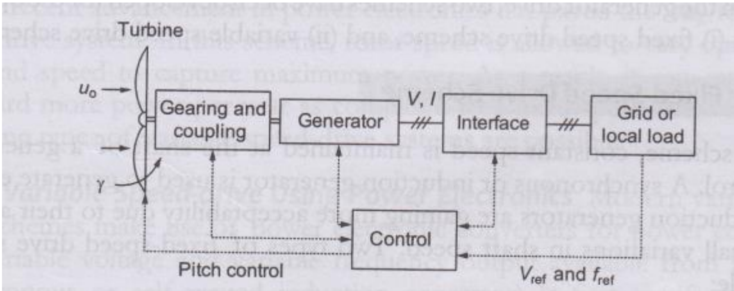
    Thermo Chemical Methods:
- Steam reforming of methane is the most energy-efficient, commercialized technology currently available and most cost effective when applied to large, constant loads.
- The method accounts for 95% of the H2 production

Electrolysis of Water

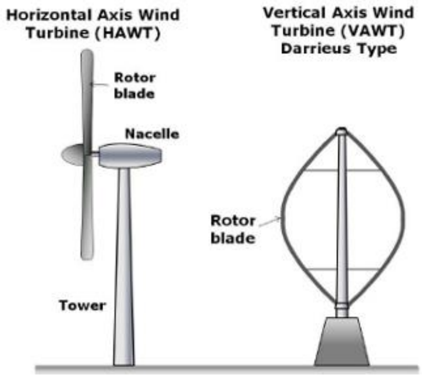
- It is the simplest method of hydrogen production.
- This method is not as cost effective as thermo-chemical method. But it generates more H2.



16 b) Explain the parts, their function and working of a wind power plant.



- The turbine shaft speed is stepped up with the help of gears, with fixed gear ratio, to suit the electrical generator & fine tuning of speed is incorporated by pitch control.
- This block acts as drive for the generator
- DC, synchronous or induction generators are used for mechanical to electrical power conversion depending on the design of the system.
- **Pitch control** is the technology used to operate and **control** the angle of the blades in a **wind turbine**.
- The system is in general either made up by electric motors and gears, or hydraulic cylinders and a power supply system.
- The interface conditions the generated power to grid quality power.
- It may consist of power electronic converter, transformer & filter etc.
- The control unit monitors & controls the interaction among various blocks.
- It derives the reference voltage & frequency signals from the grid & receives wind speed, wind direction, wind turbine etc processes them & accordingly controls various blocks for optimal energy balance.
- **1. Nacelle–** The term nacelle is derived from the name for housing containing the engines of an aircraft.
- **2. Gearbox**– The gearbox steps up the shaft rpm to suit the generator. It is the heaviest part in the nacelle.
- **3. Brakes–** Brakes are used to stop the rotor when power generation is not desired.
- **4. Generator-** It converts the energy of fast rotating shaft into electrical energy and finally the high voltage transformer converts it to high voltage to be ready to go in transmission lines.
- **5. Tower-** It's the cylindrical structure on which nacelle is mounted.



17 a) Differentiate between the performance of horizontal axis and vertical axis wind turbines. Give one examples for each type.

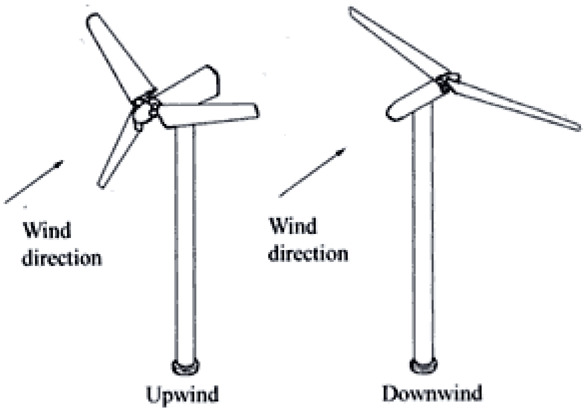
S.no	Horizontal Axis Wind Turbine	Vertical Axis Wind Turbine
1.	In HAWTs, the axis of rotation of the rotor is Horizontal to the ground.	In VAWTs the axis of rotation of the rotor is perpendicular to the ground.
2.	Yaw mechanism is present.	Absence of Yaw mechanism.
3.	It has high initial installation cost.	It has low initial installation cost.
4.	They are big in size.	They are small in size.
5.	Its efficiency is high.	It has low efficiency.
6.	It requires large ground area for installation.	It requires less ground area for installation.
7.	High maintenance cost.	Low maintenance cost as compared with HAWT.
8.	They are self-starting.	They are not self-starting.
9.	They are unable to work in low wind speed condition.	They are capable of working in low wind speed condition.
10.	Difficult in transportation.	Easy in transportation.
11.	They are mostly used commercially.	They are mostly used for private purpose only.
12.	It cannot be installed near human population.	It can be installed near human population.
13.	It is not good for the bird's population.	It is good for the bird's population.

HAWT

- It is a turbine in which the axis of rotation of rotor is parallel to the ground and also parallel to wind direction.
- They are further divided into two types
  - (i) Upwind turbine
  - (ii) Downwind turbine

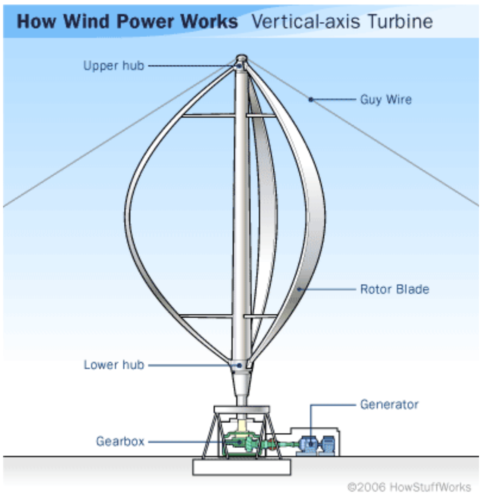
(i) Upwind Turbine

- The turbine in which the rotor faces the wind first are called upwind turbine.
- Today most of the HAWT is manufactured with this design.
- This turbine must be inflexible and placed at some distance from the tower.
- The basic advantage of this turbine is that, it is capable of avoiding wind shade behind the tower.
- It requires yaw mechanism, so that its rotor always faces the wind.
- The yaw system of wind turbines is the component responsible for the orientation of the wind turbine rotor towards the wind.



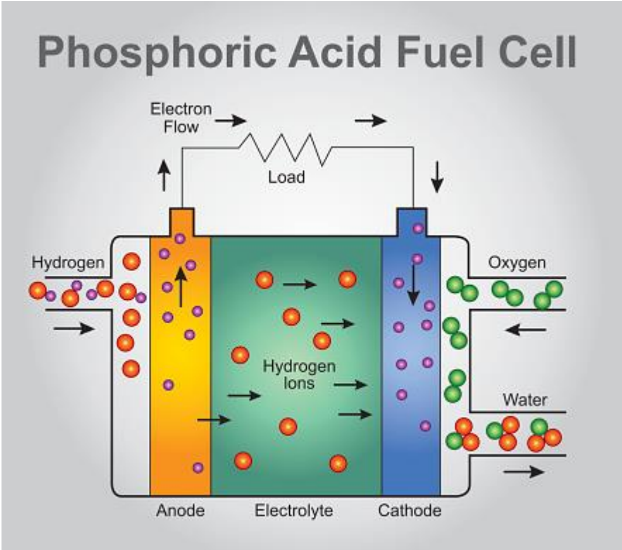
Darrieus Turbine:

- Darrieus turbine is type of HAWT. It was first discovered and patented in 1931 by French aeronautical engineer, Georges Jean Marie Darrieus. It is also known as egg beater turbine because of its egg beater shaped rotor blades.
- It consists of vertically oriented blades which are mounted on a vertical rotor. It is not a self-starting turbine and hence a small powered motor is required to start its rotation.
- First the Darrieus turbine is rotated by using a small powered motor. Once it attains sufficient speed, the wind flowing across its blades generates lift forces and this lift forces provides the necessary torque for the rotation. As the rotor rotates, it also rotates the generator and electricity is produced.



17 b) Explain the operation of a phosphoric acid fuel cell with the help of a suitable diagram.

Phosphoric acid fuel cells (PAFCs) use liquid phosphoric acid as an electrolyte—the acid is contained in a Teflon-bonded silicon carbide matrix—and porous carbon electrodes containing a platinum catalyst. The electro-chemical reactions that take place in the cell are shown in the diagram to the right. The PAFC is considered the "first generation" of modern fuel cells. It is one of the most mature cell types and the first to be used commercially. This type of fuel cell is typically used for stationary power generation, but some PAFCs have been used to power large vehicles such as city buses. PAFCs are more tolerant of impurities in fossil fuels that have been reformed into hydrogen than PEM cells, which are easily "poisoned" by carbon monoxide because carbon monoxide binds to the platinum catalyst at the anode, decreasing the fuel cell's efficiency. PAFCs are more than 85% efficient when used for the co-generation of electricity and heat but they are less efficient at generating electricity alone (37%–42%). PAFC efficiency is only slightly more than that of combustion-based power plants, which typically operate at around 33% efficiency. PAFCs are also less powerful than other fuel cells, given the same weight and volume. As a result, these fuel cells are typically large and heavy. PAFCs are also expensive. They require much higher loadings of expensive platinum catalyst than other types of fuel cells do, which raises the cost.



Phosphoric Acid Fuel Cell
Electron
Flow
Load
Hydrogen
Oxygen
Hydrogen
Ions
Water
Anode
Electrolyte
Cathode